



CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES



1. NEED FOR CLASSIFICATION

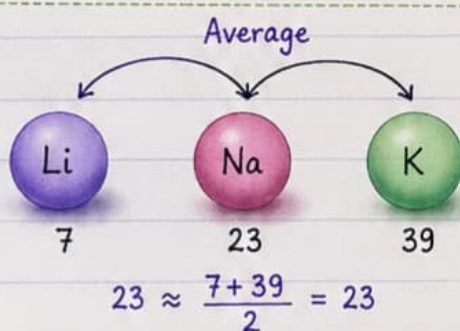
With the discovery of new elements, it became necessary to arrange them in a **systematic manner** so that their properties can be studied and compared easily.

Nature shows
Regularity,
So do the
elements!

2. EARLY ATTEMPTS AT CLASSIFICATION

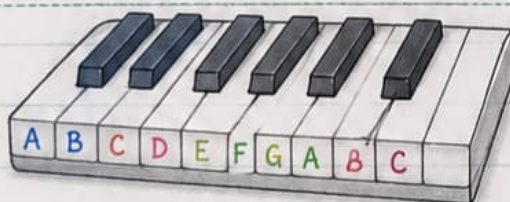
(i) Döbereiner's Triads (1829):

- Elements were grouped in threes with similar properties.
- Atomic mass of the middle element $\approx$ Average of other two elements.
- Example : Li, Na, K



(ii) Newlands' Law of Octaves (1864):

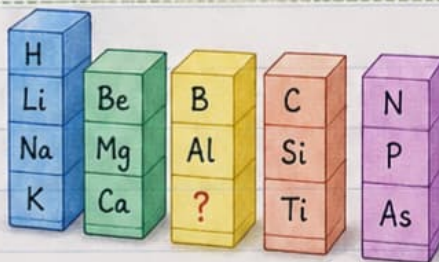
- When elements arranged in increasing order of atomic mass, every 8th element had similar properties (like musical notes).
- Limitation - Worked well only for lighter elements.



Like musical Octaves

(iii) Mendeleev's Periodic Table (1869):

- Elements arranged in increasing order of atomic mass.
- Left gaps for undiscovered elements.
- Similar properties placed in the same vertical column.
- Predicted properties of missing elements accurately.



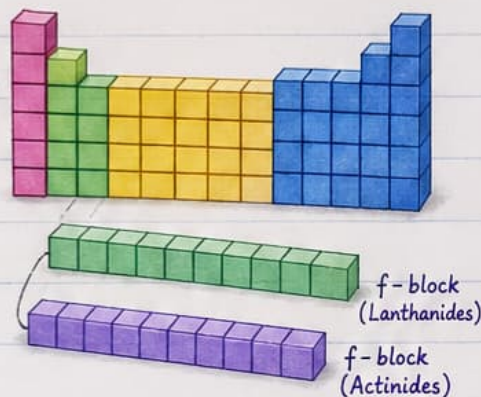
3. MODERN PERIODIC LAW



The physical and chemical properties of elements are a periodic function of their atomic numbers.

4. MODERN PERIODIC TABLE (Highlights)

- Arranged in order of increasing atomic number.
- Consists of **7 periods** (rows) and **18 groups** (columns).
- Elements in the same group have similar valence electrons.
- Divided into **s**, **p**, **d** and **f** - blocks.



5. PERIODIC TRENDS IN PROPERTIES



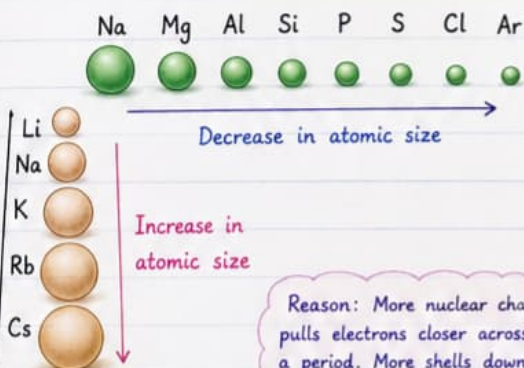
Trends occur due to changes in atomic size, nuclear charge & shielding effect.



Properties of elements show a regular (periodic) trend when we move across a period → and down a group ↓

(i) Atomic Size (Atomic Radius)

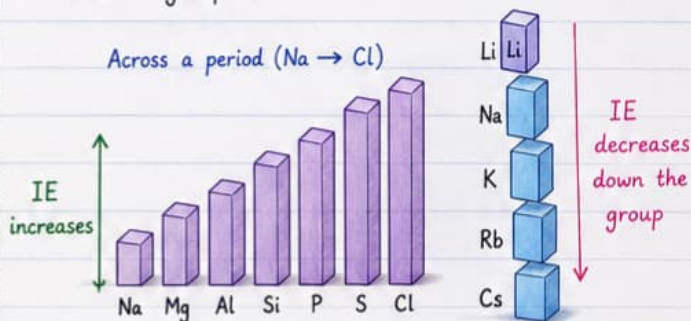
- Size of atom.
- Across a period → decreases (left to right)
- Down a group ↓ increases



Reason: More nuclear charge pulls electrons closer across a period. More shells down the group.

(ii) Ionization Enthalpy (Ionization Energy)

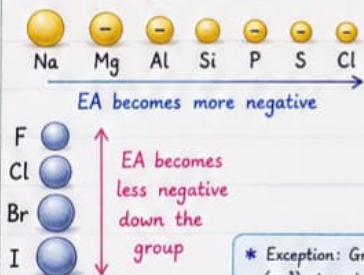
- Energy required to remove the most loosely held electron from gaseous atom.
- Across a period → increases
- Down a group ↓ decreases



Reason: Across → effective nuclear charge ↑
Down ↓ → more shells, more shielding.

(iii) Electron Affinity (EA)

- Energy released when a gaseous atom accepts an electron.
- Across a period → generally increases (becomes more negative)
- Down a group ↓ generally decreases



Reason: Across → nuclear charge ↑
Down ↓ → size ↑ so attraction for extra electron ↓

* Exception: Group 2 (ns^2) and Group 15 (np^3) show low EA due to stability.

(iv) Electronegativity (EN)

- Tendency of an atom to attract shared electron pair in a bond.
- Across a period → increases
- Down a group ↓ decreases



Reason: Across → effective nuclear charge ↑
Down ↓ → atomic size ↑, attraction ↓

6. IMPORTANT TERMS

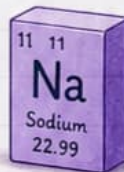
Term	Meaning
Period	Horizontal row in the periodic table. Elements in a period have same number of shells.
Group	Vertical column in the periodic table. Elements in a group have same number of valence electrons.
Valence Electrons	Electrons present in the outermost shell of an atom.
Effective Nuclear Charge (Z_{eff})	Actual positive charge felt by outermost electron after shielding.
Shielding Effect	Repulsion between inner electrons which reduces the pull of nucleus on outer electrons.



EXAMPLE (Na vs Cl)

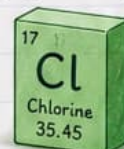
Sodium (Na)

- Larger size
- Low ionization enthalpy
- Low electronegativity
- Low electron affinity
- Highly electropositive



Chlorine (Cl)

- Smaller size
- High ionization enthalpy
- High electronegativity
- High (more negative) electron affinity
- Highly electronegative



QUICK REVISION

- ✓ Across a period (→): Size ↓, IE ↑, EA more negative ↑, EN ↑
- ✓ Down a group (↓): Size ↑, IE ↓, EA less negative ↓, EN ↓
- ✓ Trends are due to effective nuclear charge & shielding effect.
- ✓ Periodic table helps in predicting properties of elements.

Remember!

Nature follows a pattern. Once you know the trend, you can predict like a pro!



7. PERIODIC PROPERTIES (Detailed Trends)

Properties of elements change in a regular (periodic) manner with change in atomic number.



Key Idea:

Across a period → effective nuclear charge ↑

Down a group → atomic size ↑, shielding ↑



(i) Atomic Radius (Size of atom)

- Across a period → **decreases** (left to right)
- Down a group → **increases**



Decreases across a period



Increases
down a
group

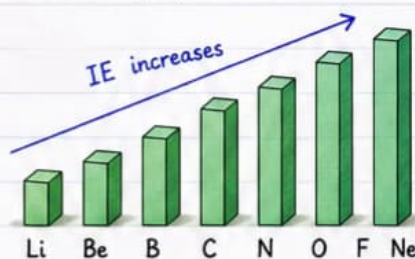
Reason:

Down the group, new shells are added → size increases.

Across a period, nuclear charge increases → pulls electrons closer.

(ii) Ionization Enthalpy (IE)

- Across a period → **increases**
- Down a group → **decreases**



IE increases



IE
decreases
down a
group

Reason:

Across → nuclear charge ↑, holds electrons tightly → more energy needed.

Down → distance ↑ and shielding ↑ → less energy needed.

(iii) Electron Affinity (EA)

- Across a period → generally **increases** (becomes more negative)
- Down a group → generally **decreases**



EA becomes more negative

F	-349
Cl	-325
Br	-295
I	-270

EA becomes
less negative
down the
group.

Reason:

Across → size ↓, nuclear charge ↑ → more attraction for extra electron.
Down → size ↑, shielding ↑ → less attraction.

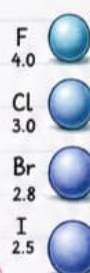
* Exception: Group 2 (ns^2) and Group 15 (np^3) show low EA due to stability.

(iv) Electronegativity (EN)

- Across a period → **increases**
- Down a group → **decreases**



EN increases



EN
decreases
down a
group

Reason:

Across → nuclear charge ↑ → more pull on shared electrons.

Down → size ↑ and shielding ↑ → less pull.



Most electronegative element → Fluorine (F)
Least electronegative → Cesium (Cs)

8. EXCEPTIONS IN PERIODIC TRENDS

Property	General Trend	Exception	Why?
IE	Increases across a period	B < Be O < N	Stability of half-filled (B: $2p^1$) and fully filled (O: $2p^4$) subshells.
EA	Becomes more negative across a period	N < O P < S	Stable half-filled subshell in N, P.
EN	Increases across a period	—	No major exception (N slightly < O)
Atomic Radius	Decreases across a period	—	No major exception
IE (Down Group)	Decreases down a group	—	No major exception

(v) Metallic Character

- Across a period → **decreases**
- Down a group → **increases**

Metallic character = tendency to lose electrons.
Metals are electropositive.



Decreases across a period



Increases
down the
group

Reason: Across → hold electrons tighter (high IE).

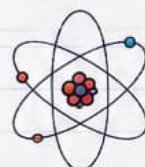
Down → easier to lose electrons (low IE).

QUICK SUMMARY

- Atomic size & metallic character → increase down group, decrease across period.
- IE & EN → increase across period, decrease down group.
- EA → becomes more negative across period, less negative down group.
- All these properties show a periodic repetition.

MEMORY TRICK

"Small Atoms, Strong Pull;
Big Atoms, Weak Pull!"



9. PERIODIC TABLE – MODERN FORM

Mendeleev's limitations were removed in the **modern periodic table**.

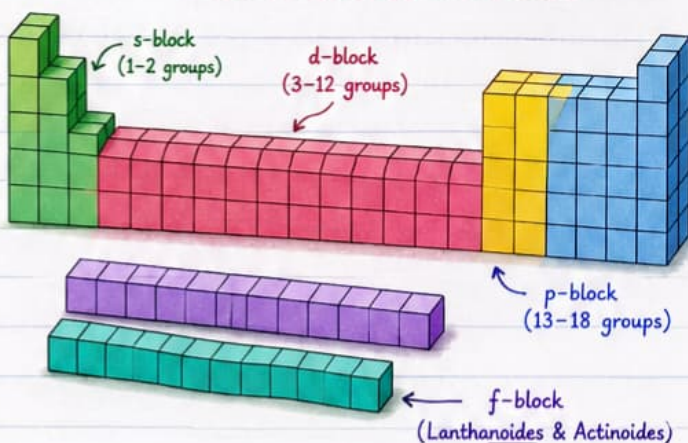
Remember:

Atomic number = Number of protons =
Number of electrons (in neutral atom)



Salient Features

- ★ Elements are arranged in order of **increasing atomic number**.
- ★ Based on electronic configuration.
- ★ Consists of 7 periods (rows) and 18 groups (columns).
- ★ Elements in the same group have similar valence electrons and chemical properties.
- ★ Divided into **s**, **p**, **d** and **f** - blocks.
- ★ Both metals and non-metals are placed systematically.
- ★ Isotopes occupy the same position.



10. VARIATION IN PROPERTIES (Across a Period)

Property	Trend Across a Period (→)	Reason
Atomic Radius	Decreases (arrow pointing right)	Effective nuclear charge ↑, pulls electrons closer.
IE Ionization Enthalpy	Increases (arrow pointing right)	Nuclear charge ↑, electrons held more tightly.
EA Electron Affinity	Generally increases (becomes more negative) (arrow pointing right)	Nuclear charge ↑, atom attracts more electrons.
EN Electronegativity	Increases (arrow pointing right)	Nuclear charge ↑, stronger pull on shared electrons.
Metallic Character	Decreases (arrow pointing right)	Atoms lose electrons with difficulty.

11. VARIATION IN PROPERTIES (Down a Group)

Property	Trend Down a Group (↓)	Reason
Atomic Radius	Increases (arrow pointing down)	New shells are added, size increases.
IE Ionization Enthalpy	Decreases (arrow pointing down)	Valence electrons are farther from nucleus, easy to remove.
EA Electron Affinity	Generally decreases (becomes less negative) (arrow pointing down)	Larger size & more shielding, less attraction for extra electrons.
EN Electronegativity	Decreases (arrow pointing down)	Atomic size ↑, attraction for shared electrons ↓.
Metallic Character	Increases (arrow pointing down)	Atoms lose electrons easily.

12. GENERAL ELECTRONIC CONFIGURATION

Block	General Electronic Configuration	Groups
s-block	ns^{1-2}	1 - 2
p-block	$ns^2 np^{1-6}$	13 - 18
d-block	$(n-1)d^{1-10} ns^{1-2}$	3 - 12
f-block	Lanthanides : $(n-2)f^{0-14} (n-1)d^{0-1} ns^2$ Actinides : $(n-2)f^{0-14} (n-1)d^{0-1} ns^2$	Lanthanides (57-71) Actinides (89-103)

- ★ Valence electrons decide the chemical behaviour of an element.



14. EXAMPLE

Why does Na ($Z = 11$) have lower IE than Mg ($Z = 12$)?

Na : $1s^2 2s^2 2p^6 3s^1$

Mg : $1s^2 2s^2 2p^6 3s^2$

Na has one valence electron in 3s orbital. It is easier to remove this one electron.

Mg has two electrons in 3s orbital, removal of one electron requires more energy.

∴ $IE(Na) < IE(Mg)$



13. IMPORTANT HIGHLIGHTS

- ★ Hydrogen is placed above Group 1, but shows properties of both Group 1 and Group 17.
- ★ Group 18 elements are called **Noble gases** or Inert gases.
- ★ Transition elements (d-block) show variable oxidation states and form coloured compounds.
- ★ Inner transition elements (f-block) are called Lanthanoids and Actinoids.
- ★ Periodic table helps in predicting physical & chemical properties and reactivity of elements.



15. MNEMONIC

Across a Period

"Size Isn't Easy

To Increase

Go And Meet

Noble Friends."

(S) Size ↓

(I) Ionization Enthalpy ↑

(E) Electron Affinity ↑

(E) Electronegativity ↑

(M) Metallic character ↓

QUICK RECAP

- ✓ Across period →
Size ↓, IE ↑, EA (more negative) ↑, EN ↑, Metallic character ↓

- ✓ Down group →
Size ↑, IE ↓, EA (less negative) ↓, EN ↓, Metallic character ↑

★ Learn the trends, Score the marks! 🍀



Understand the **pattern** today, Master the periodic table forever!

